SemantikML Evaluation Report

Generated: 2026-04-08

Compressed Evaluation Report

This PDF compresses all artifacts under evaluation/ into a single, specific summary: retrieval quality, domain-level behavior, filtered-vs-unfiltered impact, and RAG quality.

1) Evaluation Scope

- Corpus size: 9000 docs
- Domain split: research=4000, blog=3000, documentation=2000
- Retrieval evaluation queries: 25
- RAG evaluation queries: 10
- Retrieval metrics: P@5, P@10, R@5, R@10, MRR, nDCG@10, latency
- RAG metrics: Faithfulness (0-2), Relevance (0-2), latency

2) Retrieval Aggregate (All Queries)

Mode	P@5	P@10	R@10	MRR	nDCG@10	Latency
Filtered	64.8%	57.2%	84.0%	0.840	0.828	961ms
Unfiltered	61.6%	56.4%	81.5%	0.767	0.762	1448ms
Delta	+3.2pp	+0.8pp	+2.5pp	+0.073	+0.066	-487ms

- Domain filtering improves quality and speed at the same time: MRR +0.073, nDCG@10 +0.066, mean latency -487ms.

3) Retrieval by Domain (Filtered)

Domain	P@5	P@10	R@10	MRR	nDCG@10	Latency
research (n=10)	90.0%	77.0%	100.0%	1.000	0.979	1582ms
blog (n=8)	22.5%	17.5%	50.0%	0.500	0.490	383ms
documentation (n=7)	77.1%	74.3%	100.0%	1.000	1.000	734ms

- Research is near-perfect at rank quality: MRR=1.000, nDCG@10=0.979.
- Documentation is also perfect on rank position: MRR=1.000, nDCG@10=1.000.
- Blog is the weak domain: MRR=0.500, nDCG@10=0.490.

4) Query-Level Retrieval Failures (Filtered)

ID	Domain	P@10	MRR	nDCG@10	Latency
b04	blog	0.00	0.00	0.00	0.037s
b05	blog	0.00	0.00	0.00	0.036s
b06	blog	0.00	0.00	0.00	0.508s
b07	blog	0.00	0.00	0.00	0.783s
b01	blog	0.10	1.00	1.00	0.075s
b08	blog	0.10	1.00	1.00	0.066s
d01	documentation	0.10	1.00	1.00	0.526s
d04	documentation	0.10	1.00	1.00	0.147s

- Critical zeros are concentrated in blog queries: b04, b05, b06, b07.
- d04 (DataLoader) is partially covered and should be expanded in docs corpus.

5) RAG Quality

Scope	Faithfulness	Relevance	Latency

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Scope	Faithfulness	Relevance	Latency
Overall	1.70/2.00	2.00/2.00	336.087s
research	2.00/2.00	2.00/2.00	-
blog	1.00/2.00	2.00/2.00	-
documentation	2.00/2.00	2.00/2.00	-

- RAG answers are consistently on-topic (overall relevance 2.00/2.00).
- Faithfulness drops in blog content (1.00/2.00), indicating weak grounding.
- Mean generation latency is high at 336.087s/query on CPU.

6) Low-Faithfulness RAG Cases

ID	Domain	Faith	Rel	Latency
rag_b02	blog	0	2	321.713s
rag_b01	blog	1	2	191.916s
rag_d01	documentation	2	2	149.271s

- rag_b02: faithfulness=0/2, reason=Relevance: 2/3 query terms in answer (ratio=0.67). Faithfulness: 6.90% answer words overlap with context.
- rag_b01: faithfulness=1/2, reason=Relevance: 5/5 query terms in answer (ratio=1.00). Faithfulness: 12.79% answer words overlap with context.
- rag_d01: faithfulness=2/2, reason=Relevance: 4/4 query terms in answer (ratio=1.00). Faithfulness: 41.07% answer words overlap with context.

7) Root Causes and Specific Actions

- Blog corpus quality: remove duplicates and low-information templates before indexing.
- Coverage gaps: add missing DataLoader and autograd-centric docs to documentation domain.
- Retrieval policy: keep domain filtering ON by default; it improves both quality and speed.
- RAG runtime: move generation to GPU or smaller model for interactive latency targets.
- Monitoring: track per-domain MRR and faithfulness weekly to catch regressions early.

8) Artifact Inventory

- evaluation/eval_queries.json - 25 labeled retrieval queries
- evaluation/raw_query_results.json - raw top-10 hits per query
- evaluation/eval_results.json - filtered/unfiltered retrieval metrics
- evaluation/rag_eval_results.json - RAG scores and per-query outputs
- evaluation/eval_search.py - retrieval evaluation pipeline
- evaluation/eval_rag.py - RAG evaluation pipeline
- evaluation/EVALUATION_SUMMARY.md - narrative summary